

# Tide retrospective: *mixed even-power 2D moments (quartic-sextic potential)*

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## Abstract

For the 2D separable potential  $L(x, y) = x^4/24 + y^6/720$  at temperature  $t > 0$  and any  $j, k : \mathbb{N}$ , the joint even-power moment  $\langle x^{2j} \cdot y^{2k} \rangle_{2D, t}$  has the closed form

$$\left( (24/t)^{j/2} \cdot \frac{\Gamma((2j+1)/4)}{\Gamma(1/4)} \right) \cdot \left( (720/t)^{k/3} \cdot \frac{\Gamma((2k+1)/6)}{\Gamma(1/6)} \right).$$

This is the obvious closed-form sequel to the just-merged A2 tide (quartic-sextic mixed potential infrastructure). New file `Laplace/TwoD/QuarticSexticMoments.lean`, 57 lines (proof body  $\sim 8$  lines; the rest is doc-comment); zero `sorry`, zero `axiom`, zero `native_decide`. Single-attempt clean build via two `rw` calls. Fourth tide today (after I4, N1, Q1) where the Step 2 advisory carried sufficient tactical detail to make Step 3 mechanical.

## 1 Setting

A2 (`tide/quartic-sextic-mixed`, just landed) added the 2D separable-potential infrastructure for the quartic-sextic case. Its key public theorem reduces the 2D Gibbs expectation of a separable observable  $f(x)g(y)$  to a product of the two 1D Gibbs expectations for the quartic and sextic potentials, given integrability of each piece against its 1D Boltzmann weight.

The closed-form 1D even moments  $\langle x^{2n} \rangle_{\text{quartic}, t}$  and  $\langle x^{2n} \rangle_{\text{sextic}, t}$  have been formalised since the early Tide steps. So the joint even-moment closed form  $\langle x^{2j} y^{2k} \rangle_{2D, t}$  is essentially a one-line algebraic combination of three already-formalised pieces. A2's retrospective explicitly named this as the natural sequel.

The tide-pick v5 survey ranked it as the top recommendation. The Step 2 deliberation between Claude and GPT-5.5 Pro converged in a single round on Candidate A (just the closed-form combined moment); GPT additionally flagged that the existing public factor theorem already exposes the product-structure form, so a separate `pow_pow_factor` intermediate would be redundant.

The markdown tide log<sup>1</sup> carries the deliberation; the GPT consult is preserved at the sibling `gpt_responses_t1/t1_v1.md`.

## 2 The thing formalised

*For the 2D separable potential  $L(x, y) = x^4/24 + y^6/720$ , any  $t > 0$ , and any non-negative integers  $j, k$ ,*

$$\langle x^{2j} \cdot y^{2k} \rangle_{2D, t} = \left( (24/t)^{j/2} \cdot \frac{\Gamma((2j+1)/4)}{\Gamma(1/4)} \right) \cdot \left( (720/t)^{k/3} \cdot \frac{\Gamma((2k+1)/6)}{\Gamma(1/6)} \right).$$

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<sup>1</sup>In the SRI repo at `projects/primer/tide-log/`, file `2026-05-07-tide-mixed-even-2d-moments.md`.

The Lean signature<sup>2</sup>:

```
theorem gibbsExpectation_quarticSextic_pow_pow_eq
  (j k : ℕ) {t : ℝ} (ht : 0 < t) :
  gibbsExpectation
    (addSeparable Laplace.OneD.quarticPotential
      Laplace.OneD.sexticPotential) t
  (fun z : ℝ × ℝ => z.1 ^ (2 * j) * z.2 ^ (2 * k))
= ((24 / t) ^ ((j : ℝ) / 2)
  * Real.Gamma ((2 * j + 1 : ℝ) / 4)
  / Real.Gamma ((1 : ℝ) / 4))
* ((720 / t) ^ ((k : ℝ) / 3)
  * Real.Gamma ((2 * k + 1 : ℝ) / 6)
  / Real.Gamma ((1 : ℝ) / 6))
```

The 2D space is  $\mathbb{R} \times \mathbb{R}$  (per A2’s convention), not  $\text{Fin } 2 \rightarrow \mathbb{R}$ ; observables index into it via **z**.1 and **z**.2. GPT specifically called this out at deliberation as a likely point of confusion otherwise.

The closed-form RHS is kept as a literal product of the two 1D forms — no attempt to combine  $(24/t)^{j/2}$  and  $(720/t)^{k/3}$  into a single  $t$ -power. GPT flagged that combining ”creates gratuitous `rpow`/positivity/algebra overhead” with no mathematical or readability gain.

### 3 Proof strategy

The proof is two `rw` calls:

1. Apply A2’s separable-observable factor theorem with  $f := x \mapsto x^{2j}$  and  $g := y \mapsto y^{2k}$ , using the existing 1D integrability-of-monomial-against-Boltzmann-weight witnesses for the quartic and sextic potentials. The 2D expectation becomes a product of two 1D expectations.
2. Substitute the 1D even-moment closed forms (one quartic, one sextic) to obtain the Gamma-form RHS.

The proof body is eight lines of Lean.

### 4 Roadblocks and resolutions

*None.* Single-attempt clean build, zero warnings. This is the fourth tide today (after I4, N1, Q1) where a clean Step 2 advisory made Step 3 mechanical. The pattern is now reliable: when the Step 2 deliberation correctly identifies the integrability bundle, the namespace conventions, and the verbatim form of the closed-form RHS, the proof writes itself.

**GPT-5.5 Pro:** Big Lean caveat: copy the RHS verbatim from the 1D theorems. In particular, match exactly whether they use `^` or `Real.rpow`, `((2 * j + 1 : ℝ) / 4)` vs extra casts, parenthesisation of `*` and `/`. If you match the 1D theorem statements exactly, the proof is basically two rewrites. If you ”prettify” the RHS, you’ll buy yourself avoidable algebra.

The implementation took this guidance literally — the RHS is character-for-character a product of the two 1D theorem RHSes. The proof is two rewrites.

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<sup>2</sup>`Laplace/TwoD/QuarticSexticMoments.lean`, theorem `gibbsExpectation_quarticSextic_pow_pow_eq`.

**Numerical sanity pre-Step 3.** Eight cases at  $t \in \{1, 2\}$ ,  $(j, k) \in \{0, 1, 2\}^2$ , all agree to  $\sim 10^{-13}$  relative error against `scipy.integrate.dblquad`. The identity is exact (a structural separability fact for the 2D measure), not asymptotic.<sup>3</sup>

## 5 What was Mathlib, what was new

**Mathlib provided.** Nothing directly — this tide is entirely above the Mathlib stack, composing already-formalised Laplace/laplace pieces.

**What was new.** A single public theorem (`gibbsExpectation_quarticSextic_pow_pow_eq`) that combines three already-formalised pieces:

- A2’s separable-observable factor theorem (the 2D-to-1D reduction);
- the existing 1D integrability witnesses for monomials against the quartic and sextic Boltzmann weights;
- the 1D even-moment closed forms for the quartic and sextic potentials.

The novelty is just the composition; no new lemma machinery.

## 6 Lessons

- *Sometimes the right Tide is the trivial composition.* A2 just landed; the obvious sequel is a 30-line composition theorem. There is no value-add in proving anything more elaborate in this Tide step. Resist the impulse to bundle further generalisations (odd-power vanishing, generic pure-monomial pairs, etc.) — those are separate Tide steps.
- *The "match RHS verbatim" rule.* Whenever a Tide composes two existing theorems with literal product-of-RHS targets, write the headline RHS character-for-character as the product of the two 1D RHSes. This makes the proof a pure `rw` chain. Any algebraic re-expression — even a *semantically equivalent* one — will introduce avoidable `rpow` and positivity steps.
- *The "is the existing public theorem already enough?" test.* GPT correctly noted that proposing a separate `pow_pow_factor` public intermediate (Candidate B) would be redundant given A2’s existing `*_factor` theorem already exposes the product structure for arbitrary  $f, g$ . When a Tide proposes a sub-result that’s already a special case of an existing public theorem, the right answer is "the existing theorem is enough; just show how to use it". Worth recording for future Tide deliberations.

## 7 Follow-ups

- *Mixed odd-power vanishing* (Candidate C, deferred):  $\langle x^{2j+1}y^{2k} \rangle_{2D,t} = 0$  and the symmetric variant. Each follows from the factor theorem plus the existing 1D odd-moment vanishing theorems. About  $\sim 30$  lines for the parity-completion pair.

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<sup>3</sup>Script preserved at `projects/primer/tide-log/gpt_responses.t1/t1_numerics.py`.

- *Other 2D mixed pure-monomial pairs.* Quartic-quartic, sextic-sextic,  $x^{2j}/((2j)!)$  paired with  $y^{2k}/((2k)!)$  for arbitrary  $(j,k)$ . Each is a 30-line corollary of A3 once the named 1D pure-monomial families exist; A2’s retrospective named this as a follow-up. Best deferred until A1 (the generic even-monomial 1D template) lands.
- *Generic- $k$  J-function-style 2D moment asymptotic.* The natural *asymptotic* sequel: as  $t \rightarrow \infty$ , the 2D moment behaves as the product of the leading-order 1D scalings. Quartic side scales as  $t^{-(2j+1)/4}$ , sextic side as  $t^{-(2k+1)/6}$ , joint scaling  $t^{-(2j+1)/4-(2k+1)/6}$ . Tide-shaped;  $\sim 80$  lines combining `kth_jfunction_asymptotic` machinery from N1.
- *\leanref-tag the primer’s 2D examples* once the primer has TeX statements for any of the 2D moment formulas. Pin to this tide’s commit. Process tide; pairs with K2.

## Acknowledgements

GPT-5.5 Pro contributed the deliberation-stage advisory at Step 2: the verdict for Candidate A over B (B is redundant given A2’s existing public factor theorem) and C (better as a separate parity-completion follow-up); the `z.1/z.2` naming guidance; and the verbatim-RHS rule that made the proof a pure two-line `rw` chain. The full consult is preserved alongside the markdown tide log.<sup>4</sup> The proof landed single-attempt with no mid-proof thrashing-rescue consult needed.

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<sup>4</sup>In the SRI repo at `projects/primer/tide-log/gpt_responses_t1/`, file `t1_v1.md`.